There are three types of consumable electrodes. They are

1. Bare electrodes
2. Lightly coated electrodes
3. Heavily coated electrodes

Non – consumable electrodes are used in the processes of atomic hydrogen welding and TIG welding.

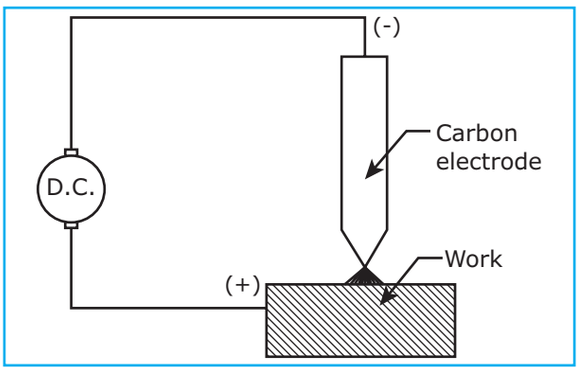
7.9 Selection of electrodes

The material used for manufacturing electrodes depends upon the material to be welded. Given is the list indicating suitable materials for manufacturing electrodes for welding different metals.

S. No	Material to be welded	Electrode material
1	Wrought iron	Low Carbon Steel Rod
2	Mild Steel	Mild Steel, Copper coated Rod
3	Alloy Steel	Nickel Steel Rod
4	Cast Iron	Cast Iron Rod
5	Aluminium	Cast Aluminium Alloy Rod
6	Carbon Steel	Steel wire (0.15% Carbon & 0.025%) Phosperous & Sulphur
7	Copper	Copper Rod
8	Brass	Brass Rod

7.10 Carbon arc welding

In carbon arc welding, the process of welding is carried out by an electric arc. The arc formed between the electrode and the work piece generates high amount of heat. In D.C electric supply, the carbon electrode is connected to the negative terminal and the positive terminal is connected to the work piece. During the formation of the electric arc, the temperature of the positive terminal is 4000^0C and the negative terminal will be around 3000^0C. Carbon, the electrode material will not fuse with the work piece materials. This prevents the joint from becoming weak. A lengthy arc produces carbon – monoxide at the location of welding and prevents the molten metal from reacting with the atmospheric air.



Carbon arc welding

Both ferrous and non- ferrous metals can be welded by this method. Steel sheets, copper and its alloys, brass and aluminium parts are welded using carbon arc welding. This method of welding can also be done automatically. Starting of the electric arc is easy and the temperature is controlled easily. The disadvantage of this method is the presence of blow holes in the joint.

7.11 Gas welding

Gas welding is the process of melting and joining metal parts by means of a gas flame. Generally pressure is not applied during the process of gas welding. Oxygen and acetylene gases are made to pass through the welding torch. These gases are mixed at the required ratio

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